Parallel qMRI Reconstruction from 4x Accelerated Acquisitions

WashU

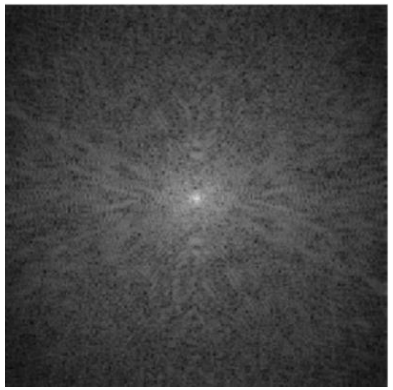
James McKelvey School of Engineering

Mingi Kang, Ulugbek S. Kamilov PhD, Shirin Shoushtari

Computational Imaging Group, Washington University in St. Louis

MRI Acquisitions

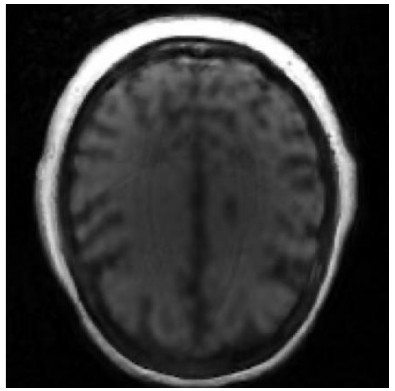
K-Space (Complex)



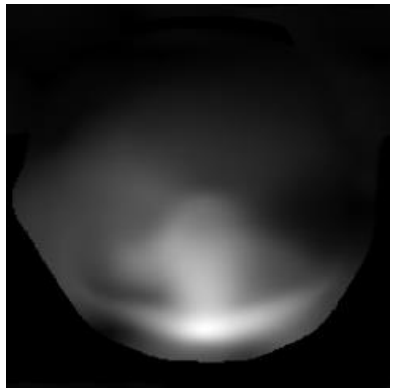
F^{-1}

F

Image Space (Real)



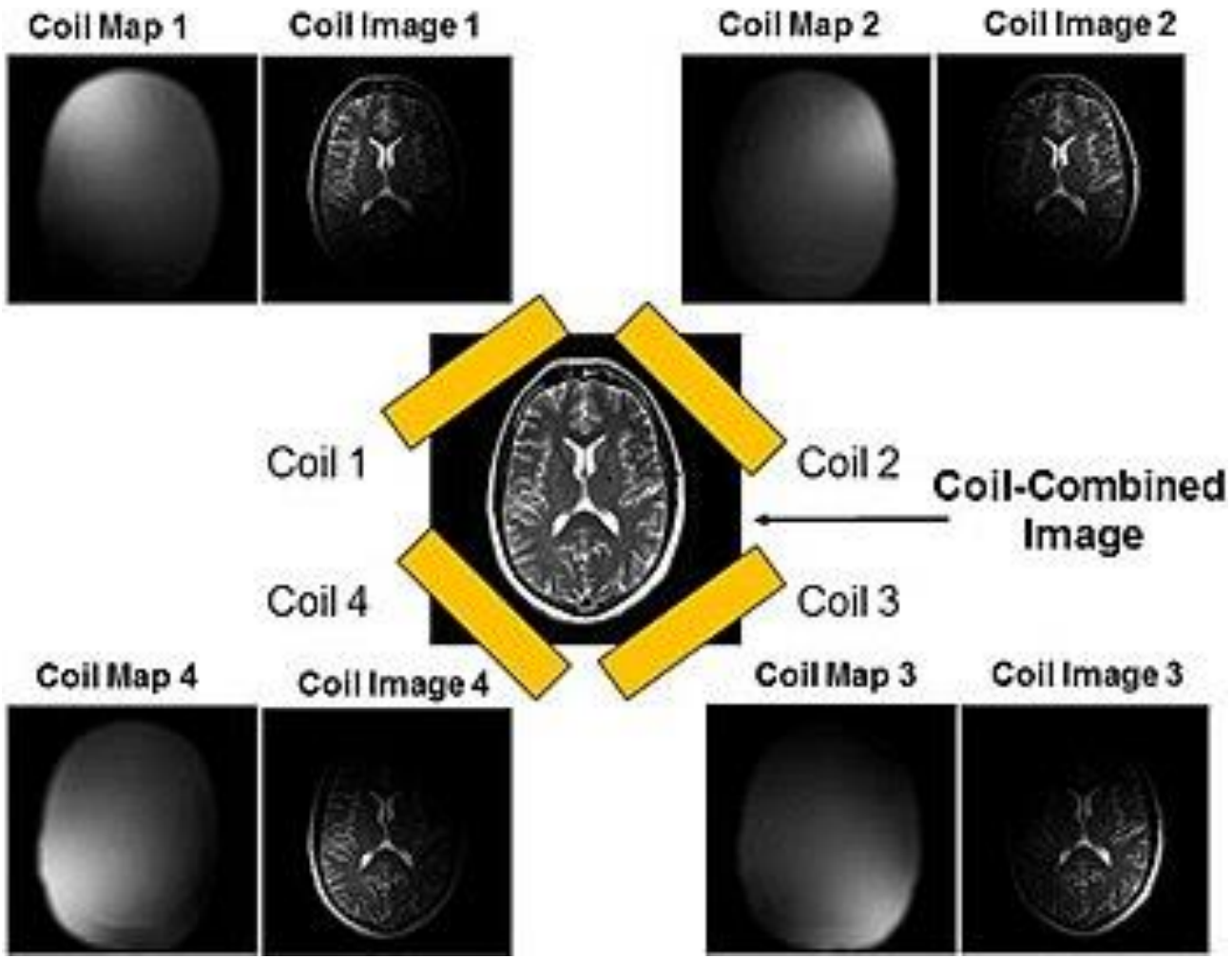
Coil Sensitivity Maps



F = Fourier Transformation
Frequency Domain ⟷ Image Domain

https://wtop.com/health-fitness/2022/02/new-mri-helps-patients-with-implanted-devices-claustrophobia-or-obesity/

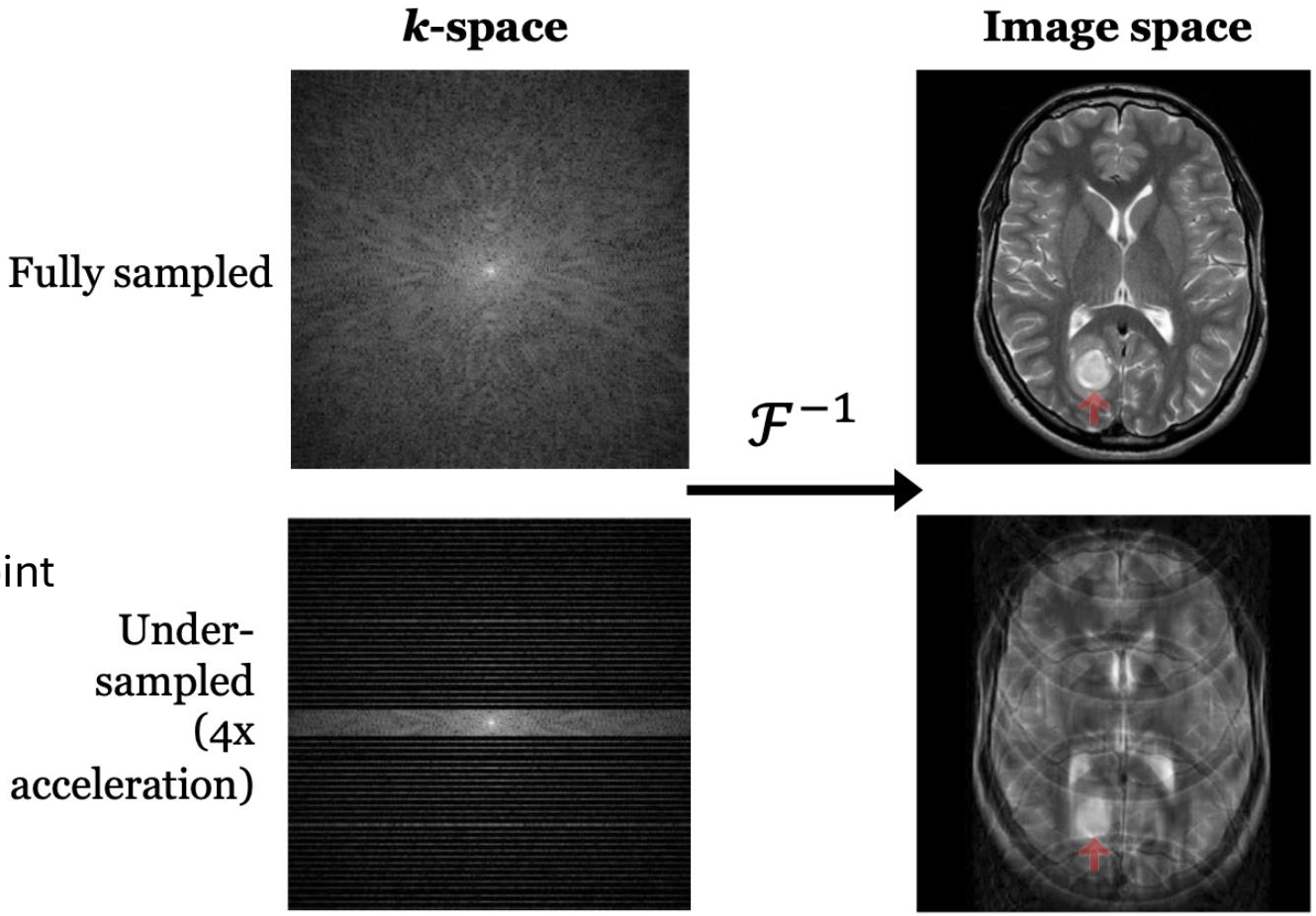
Coil Sensitivity Maps



https://en.wikipedia.org/wiki/Real-time_MRI

Accelerated MRI Acquisition

4x Acceleration
=
measure every 4th line/point



V. Prasad, R. J. G. van Sloun, S. v. d. Elzen, A. Vilanova and N. Pezzotti, "The Transform-and-Perform Framework: Explainable Deep Learning Beyond Classification," in IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 2, pp. 1502-1515, Feb. 2024, doi: 10.1109/TVCG.2022.3219248.

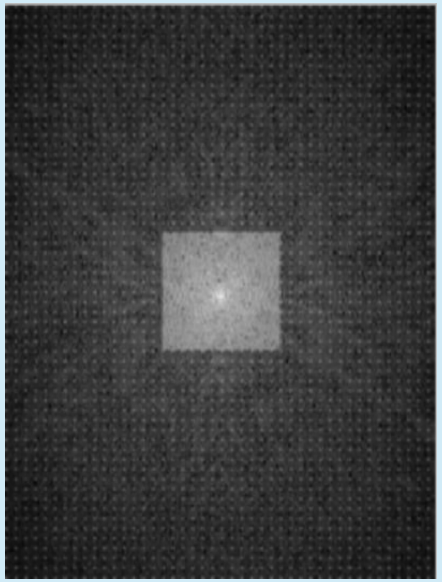
Real MRI Data from WashU Medical School

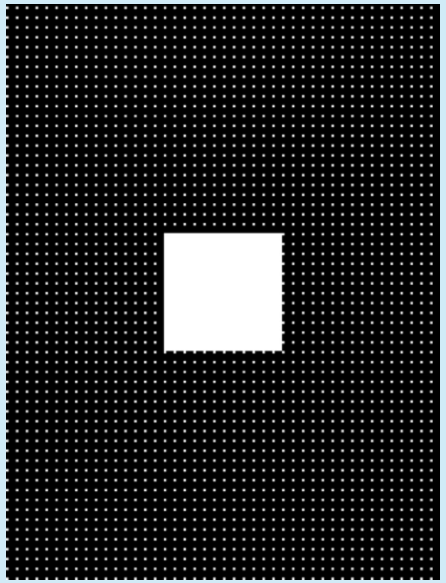
Total of 5,600 MRI Images for 2x and 4x Accelerated Acquisitions

- From 8 patients
- Brain MRI Images

Accelerated MRI Acquisition

4x Accelerated





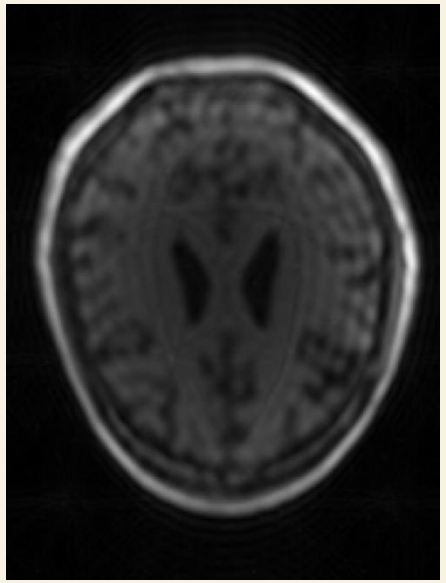
Visual of sampling pattern for 4x accelerated

Apply

Zero Filling
(Fill empty with zero)

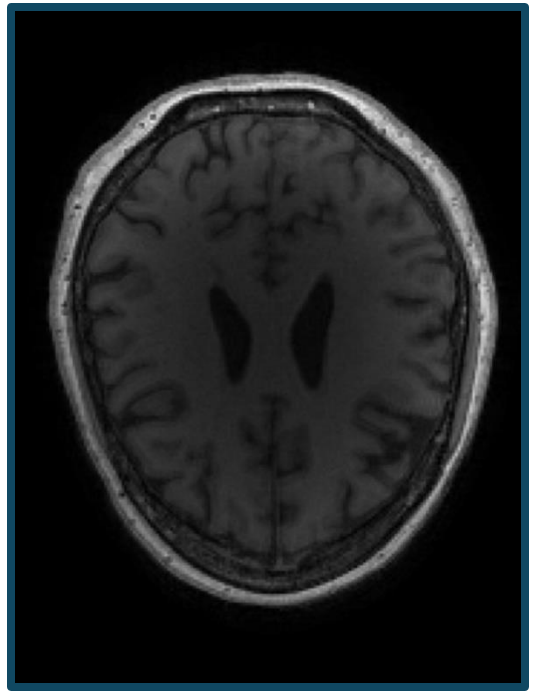
F^{-1}

Pre-Reconstructed Image



Inconsistency between 2x and 4x Acceleration

2x Acceleration

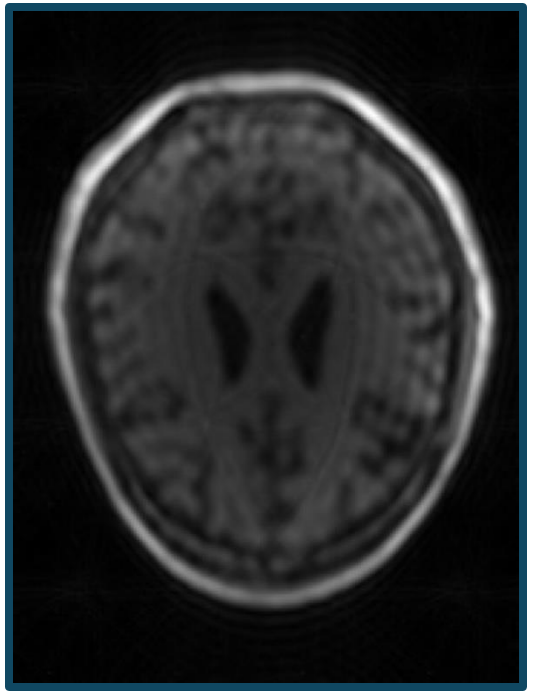


$\neq$

Overall different images

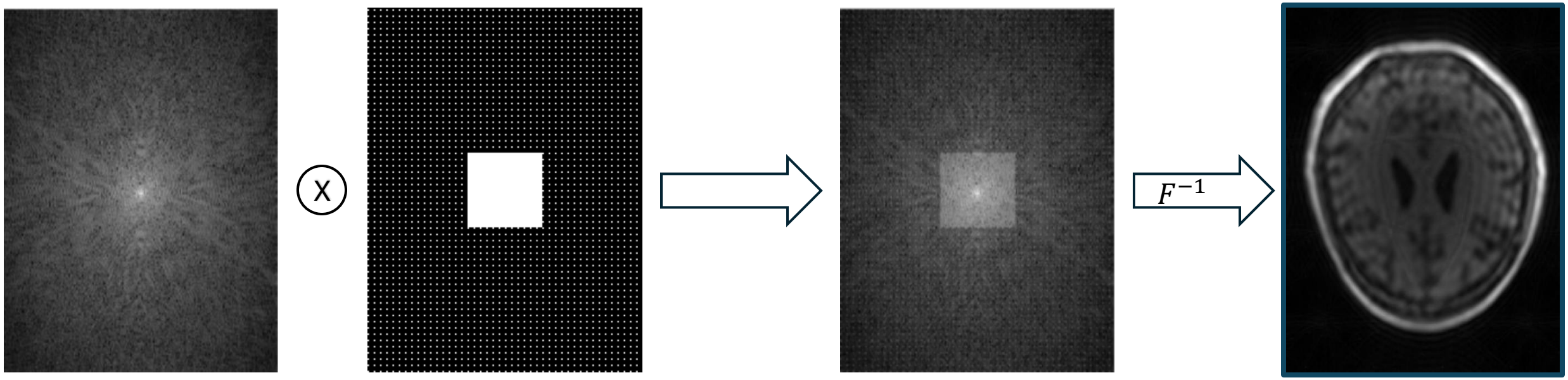
Physical shifts between 2x and 4x measurements

4x Acceleration



Consistent 2x and 4x pairs for Training

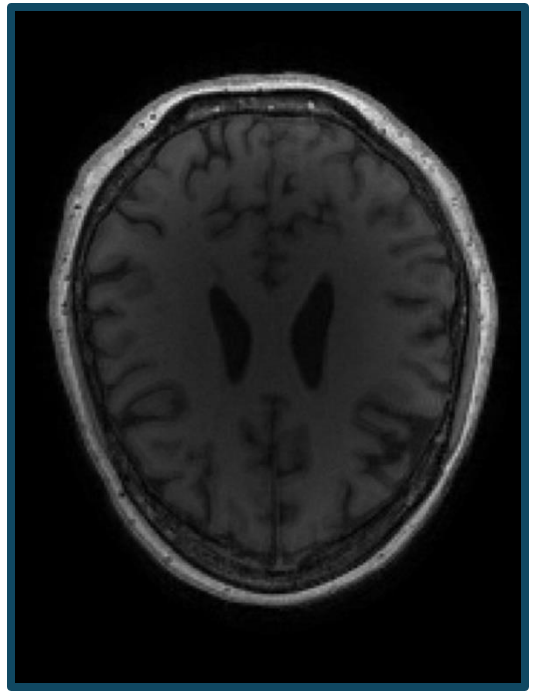
2x Ksp **4x Mask** **Simulated 4x Ksp** **Simulated 4x Img**



Mask is applied to full K-Space then Inverse Fourier transform to image space

Consistent 2x and 4x pairs for Training

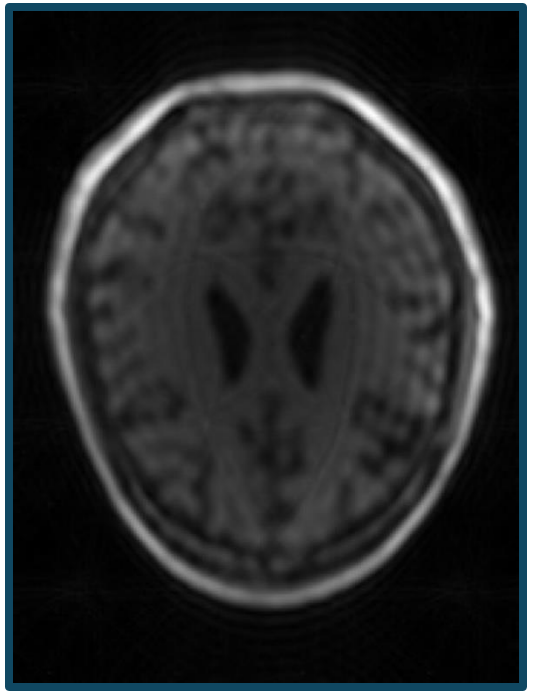
2x Acceleration



Same image, different quality

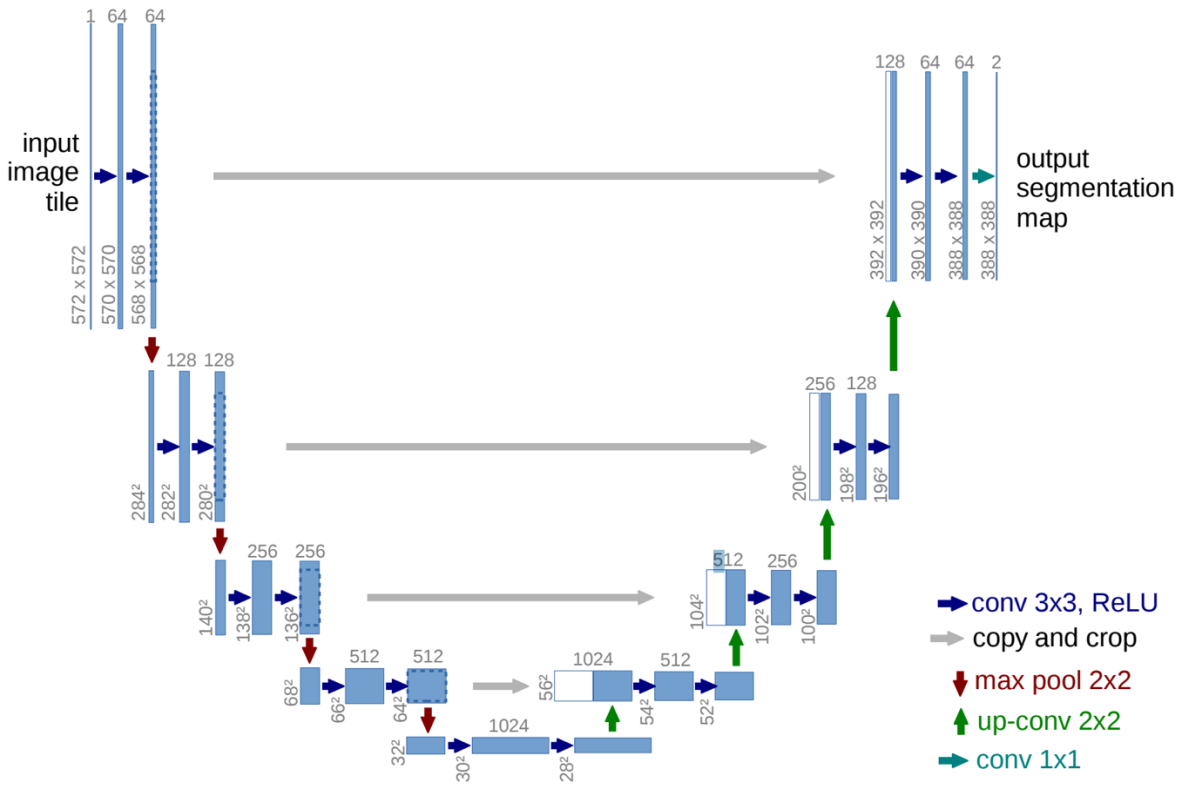
Consistent for training

Simulated 4x Acceleration



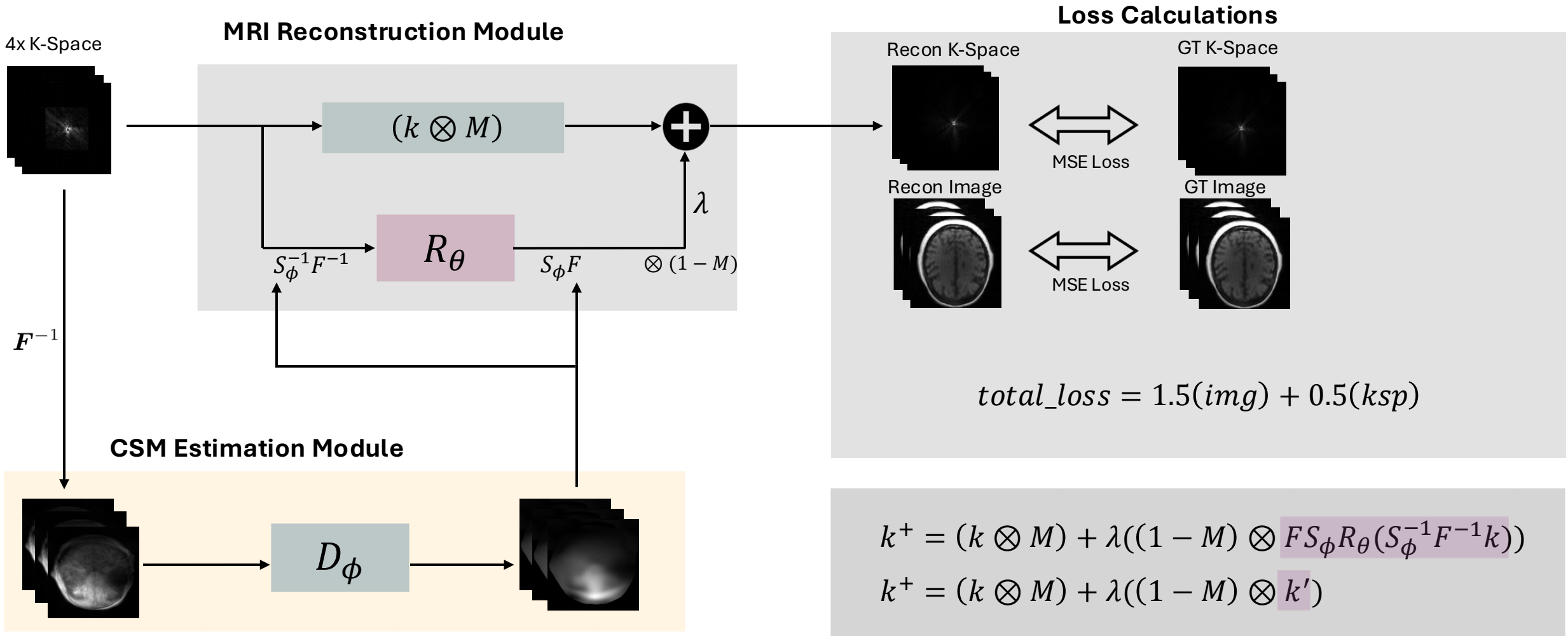
Model Architecture

U-Net: Convolutional Networks for Biomedical Image Segmentation



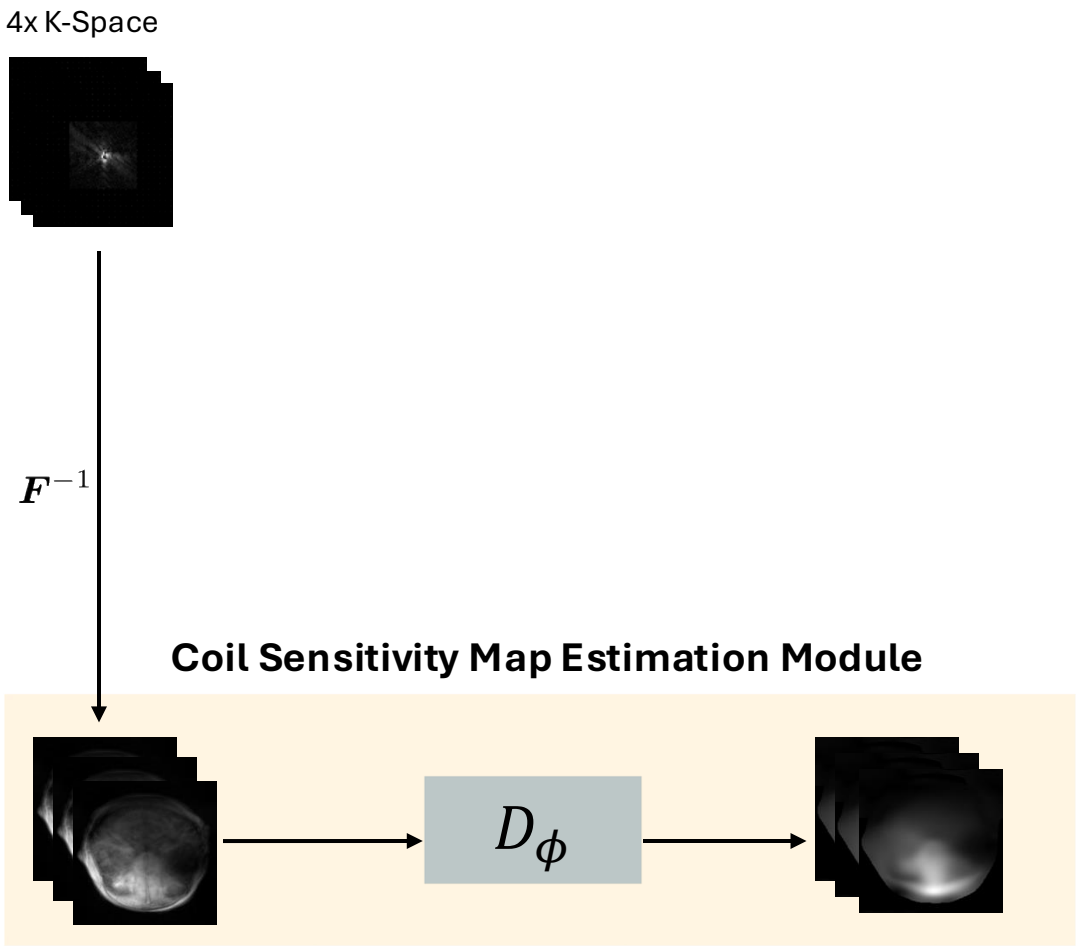
Ronneberger, O., Fischer, P., Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In: Navab, N., Hornegger, J., Wells, W., Frangi, A. (eds) Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. MICCAI 2015. Lecture Notes in Computer Science(), vol 9351. Springer, Cham. https://doi.org/10.1007/978-3-319-24574-4_28

Training Pipeline

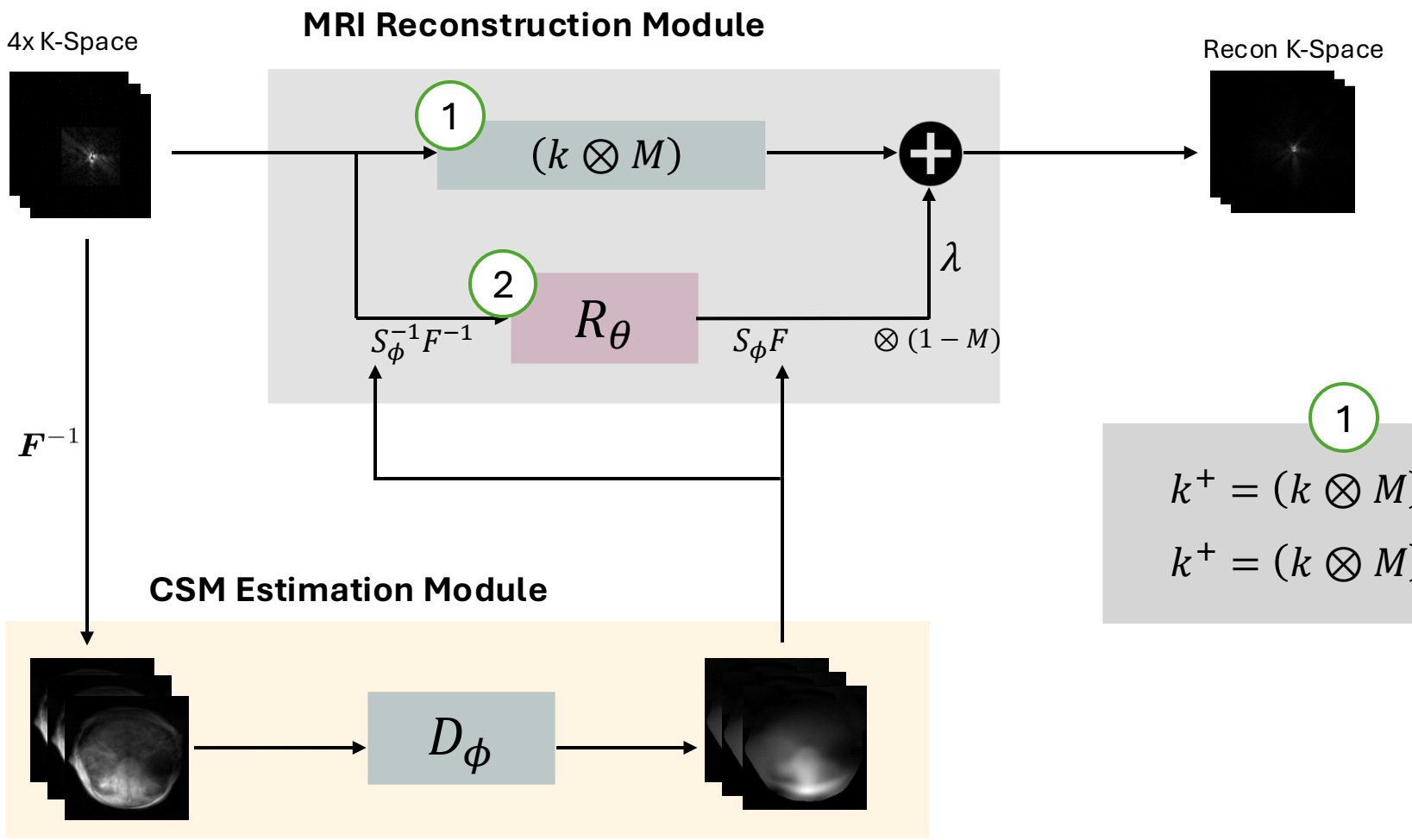


Hu Y, Gan W, Ying C, Wang T, Eldeniz C, Liu J, Chen Y, An H, Kamilov US. SPICER: Self-supervised learning for MRI with automatic coil sensitivity estimation and reconstruction. Magn Reson Med. 2024 Sep;92(3):1048-1063. doi: 10.1002/mrm.30121. Epub 2024 May 10. PMID: 38725383.

Coil Sensitivity Estimation Model



MRI Reconstruction Module



1 2

$$k^+ = (k \otimes M) + \lambda((1-M) \otimes FS_\phi R_\theta(S_\phi^{-1}F^{-1}k))$$

$$k^+ = (k \otimes M) + \lambda((1-M) \otimes k')$$

Pipeline in Equation Form

$$k^{+} = (k \otimes M) + \lambda((1 - M) \otimes FS_{\phi}R_{\theta}(S_{\phi}^{-1}F^{-1}k))$$

$$k^{+} = (k \otimes M) + \lambda((1 - M) \otimes k')$$

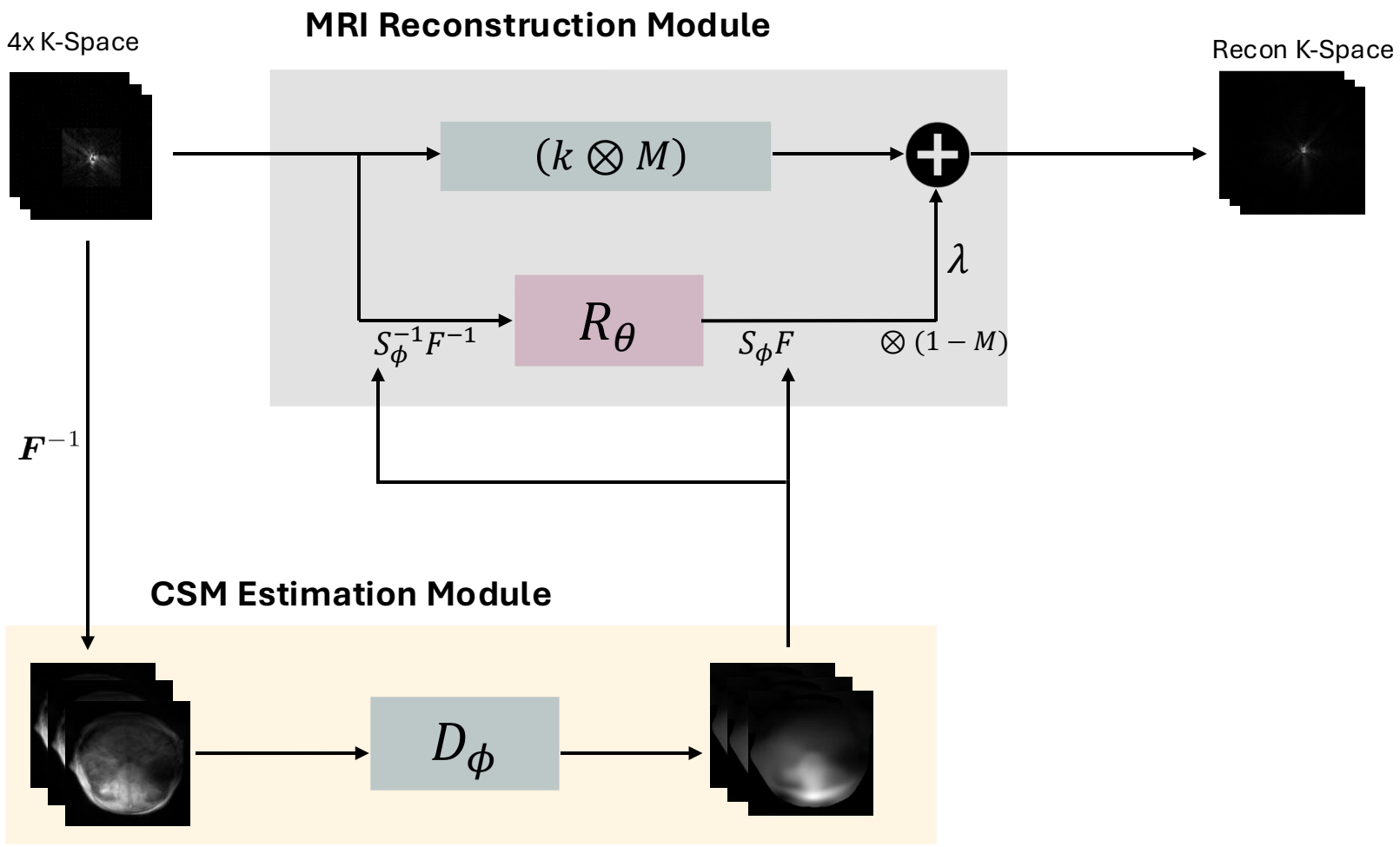
k^{+}= Final Reconstructed K-Space Output

$(k \otimes M)$ = Keep originally sampled K-Space

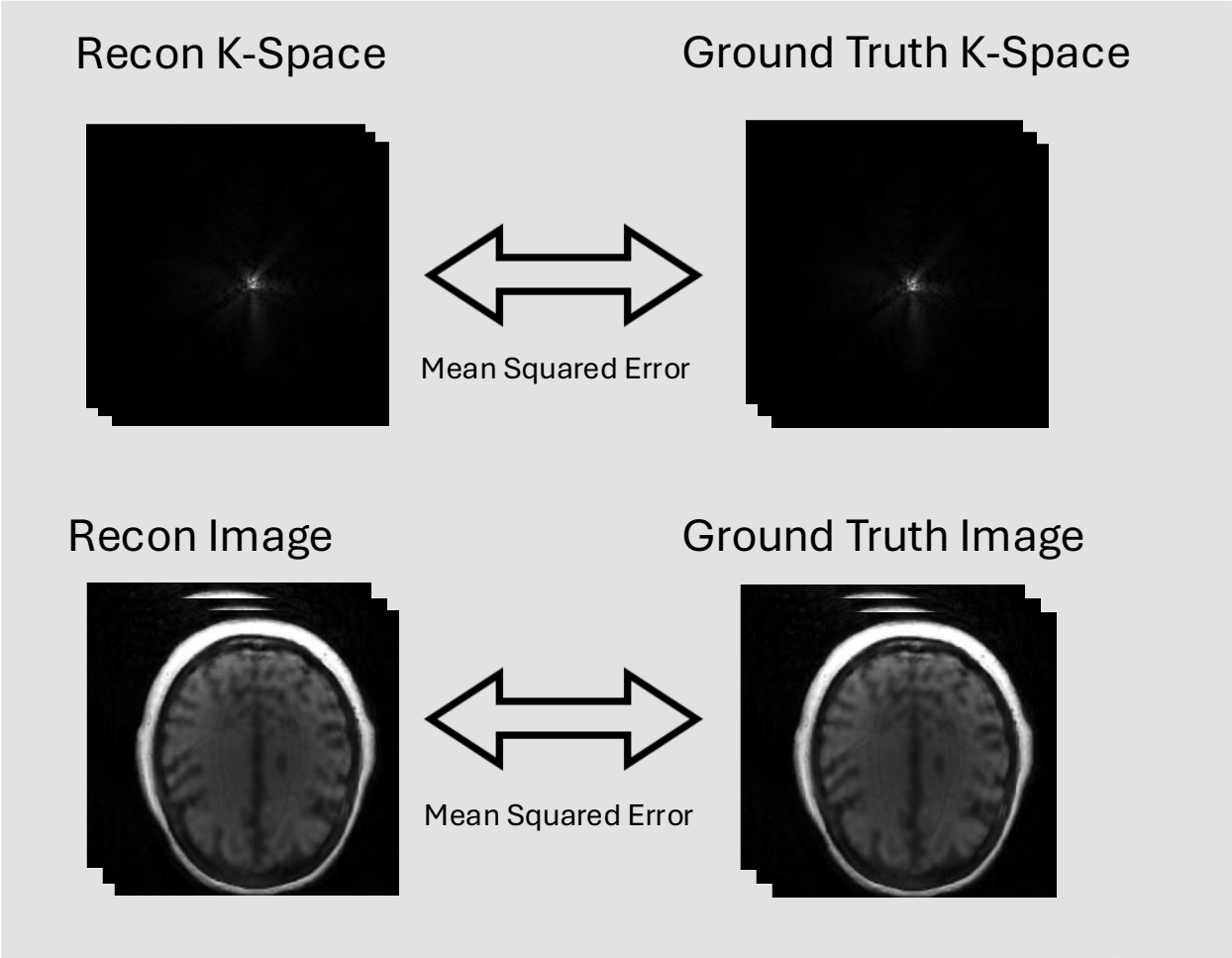
$k' = FS_{\phi}R_{\theta}(S_{\phi}^{-1}F^{-1}k))$ = Reconstructed Output K-Space from the Model

$\lambda((1 - M) \otimes k')$ = Only take missing positions from Reconstructed Output

MRI Reconstruction Module



Loss Calculations



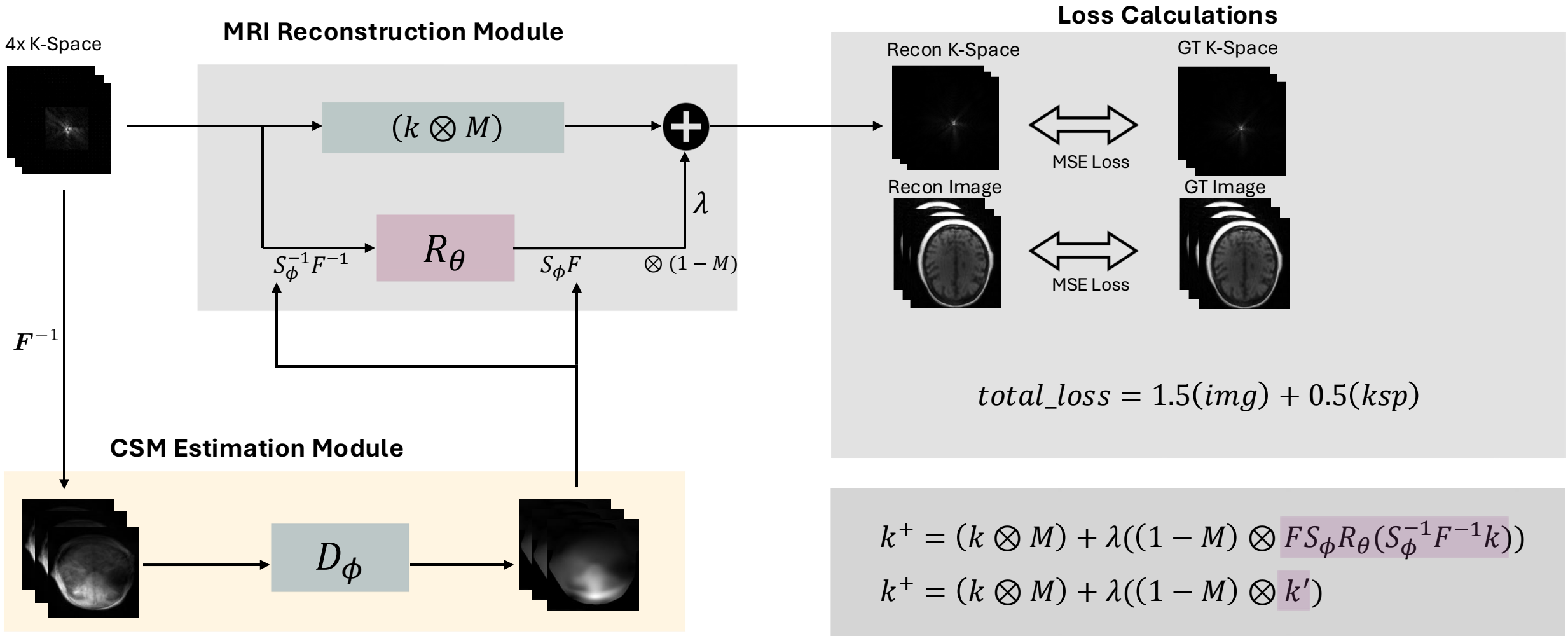
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y^{+} - y^{gt})^2$$

y^{+}= Output of the Model (Reconstructed)

y^{gt}= Full Reference Data (Ground Truth)

$$total_loss = 1.5(img) + 0.5(ksp)$$

Training Pipeline



Hu Y, Gan W, Ying C, Wang T, Eldeniz C, Liu J, Chen Y, An H, Kamilov US. SPICER: Self-supervised learning for MRI with automatic coil sensitivity estimation and reconstruction. Magn Reson Med. 2024 Sep;92(3):1048-1063. doi: 10.1002/mrm.30121. Epub 2024 May 10. PMID: 38725383.

Measuring Reconstruction Performance

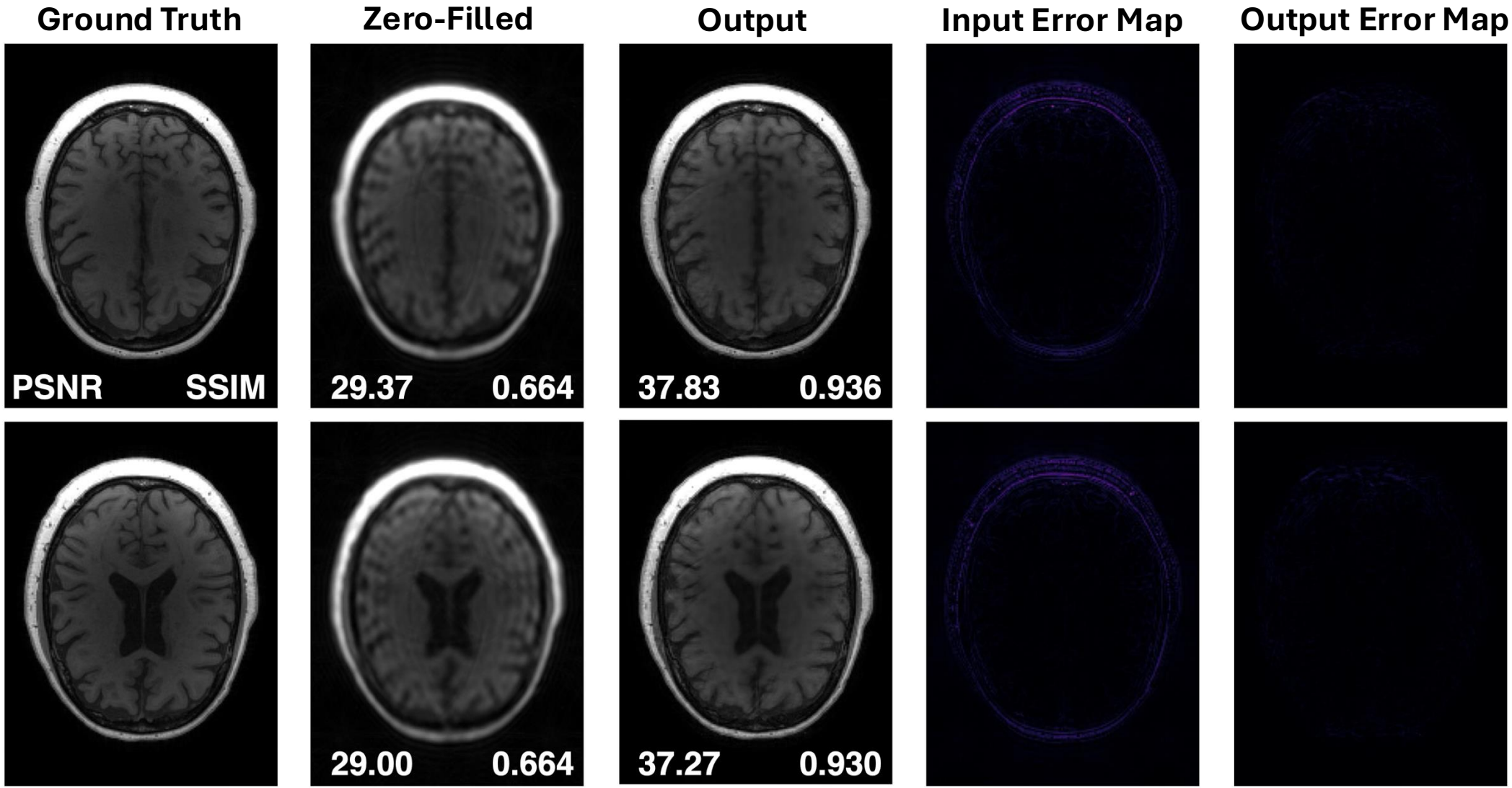
PSNR – Peak Signal-to-Noise Ratio

- Measures ratio between maximum power of a signal and maximum power of noise
- Measured in decibels (dB).
- **Higher PSNR = Better Reconstruction Quality**

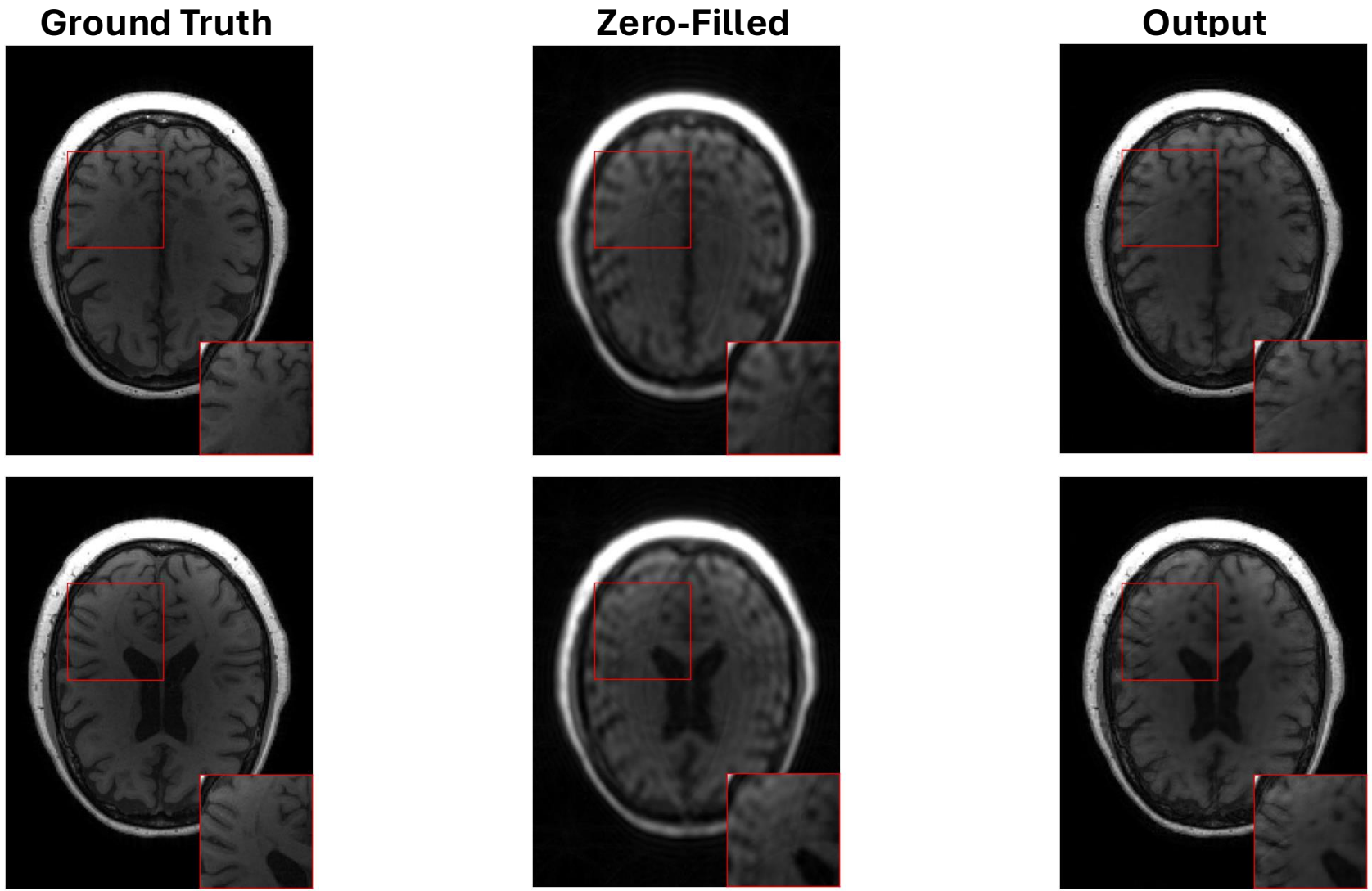
SSIM – Structural Similarity Index Measure

- Measures perceptual similarity between two images with changes in luminance, contrast, and structure
- Range of (0, 1)
- **SSIM closer to 1 = Better Reconstruction Quality**

Train & Test on Simulated 4x



Train & Test on Simulated 4x

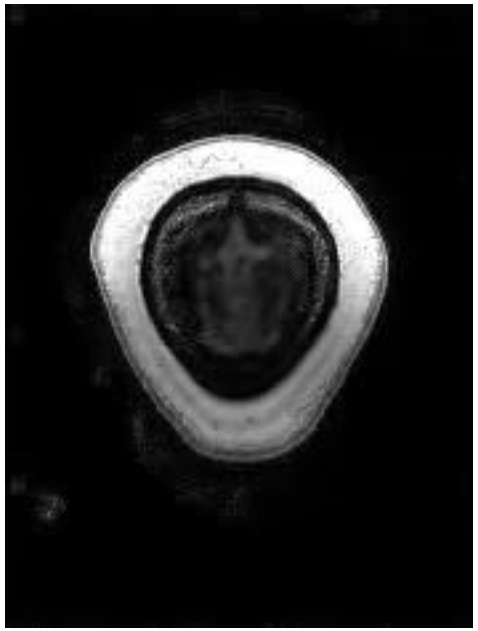


Remark: Error Maps scaled by 5 for visualization

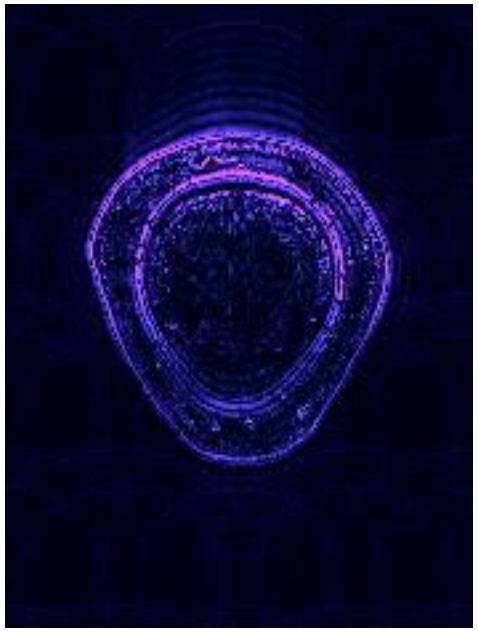
Ground Truth

Zero-Filled

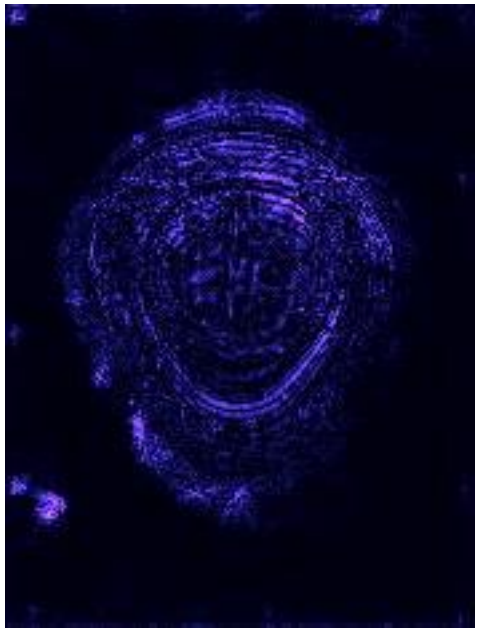
Output



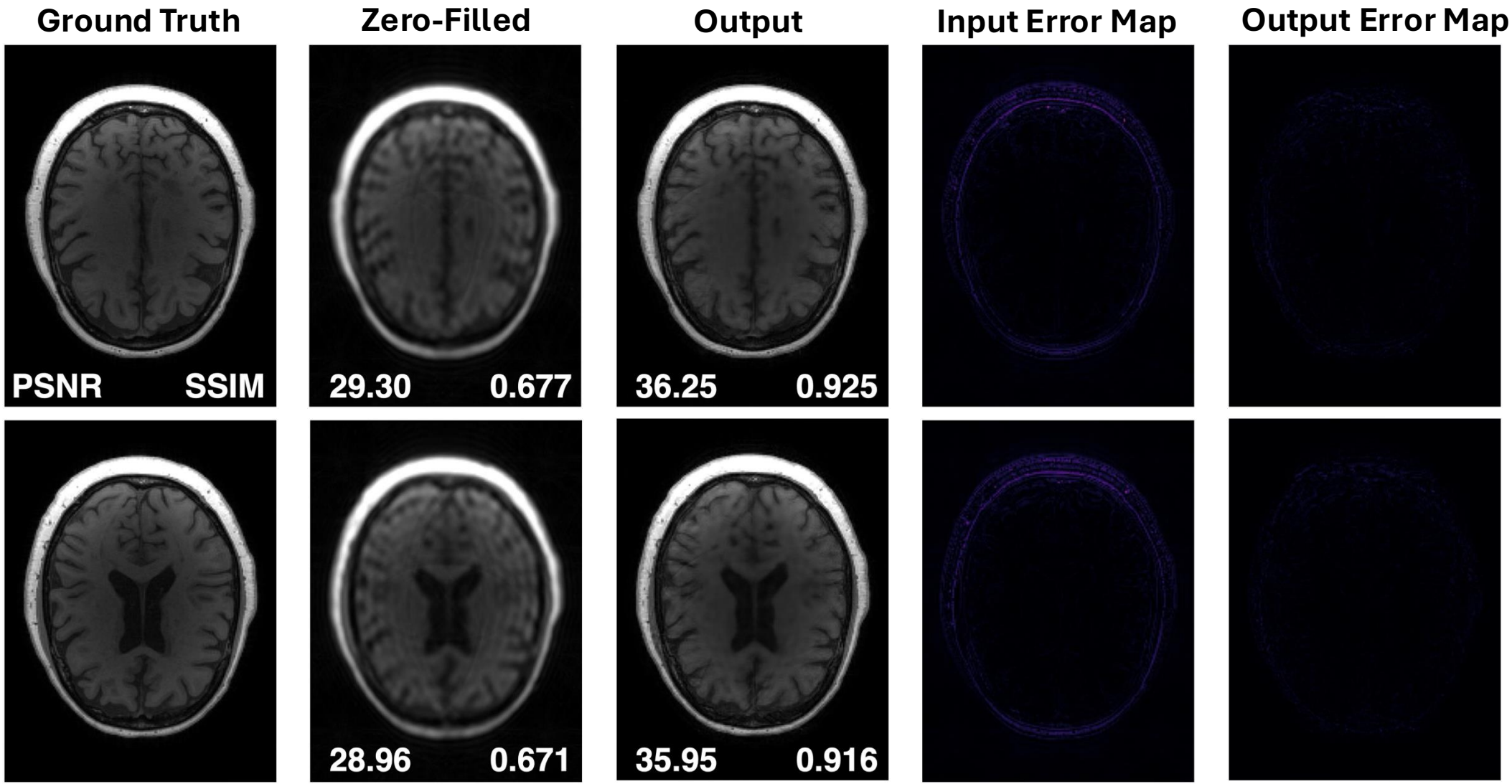
Input Error Map



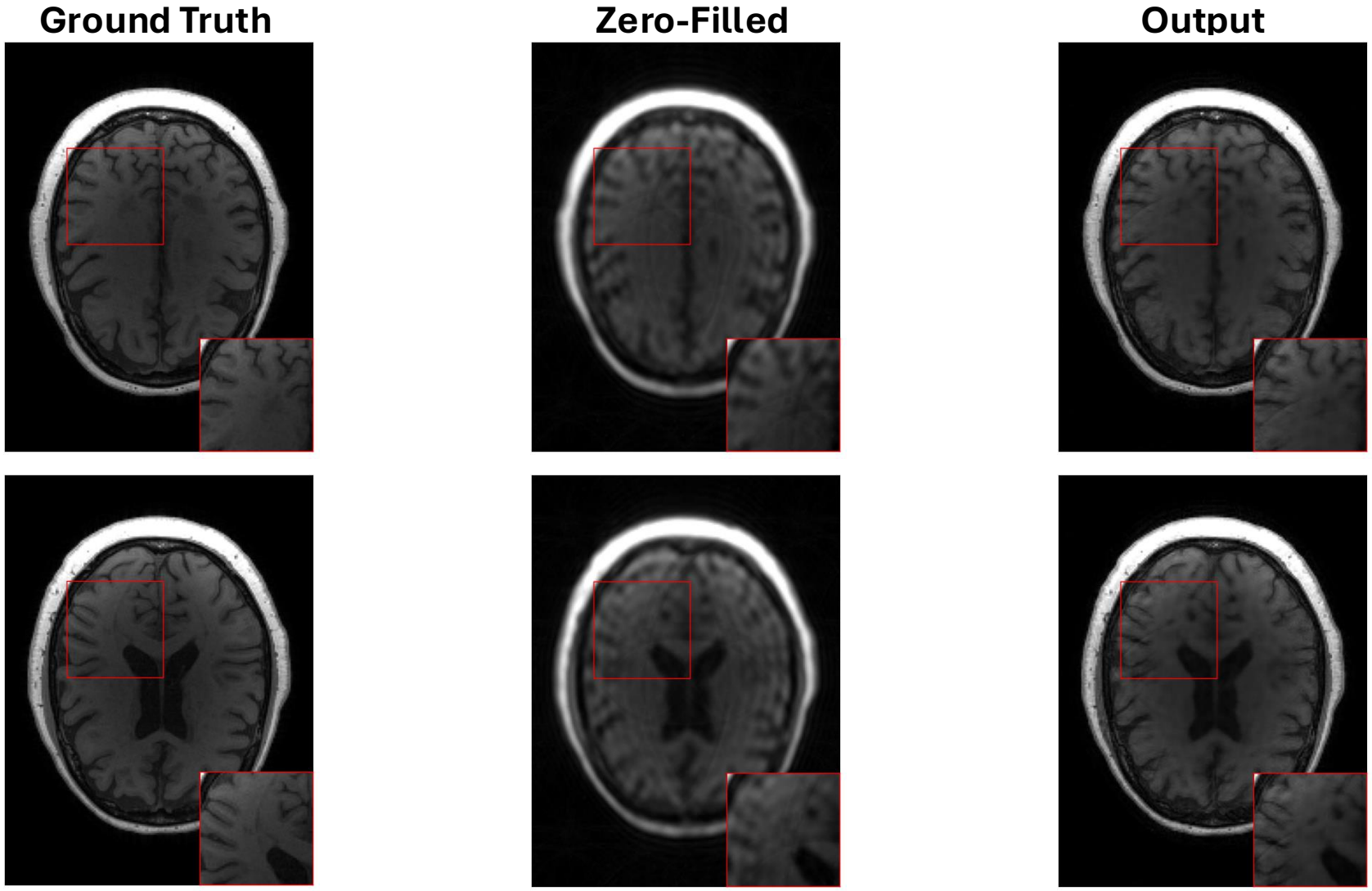
Output Error Map



Train on Simulated 4x & Test on Real 4x



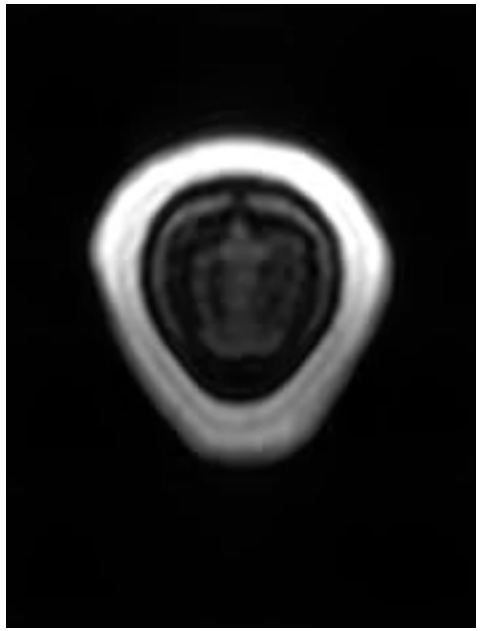
Train on Simulated 4x & Test on Real



Remark: Error Maps scaled by 5 for visualization

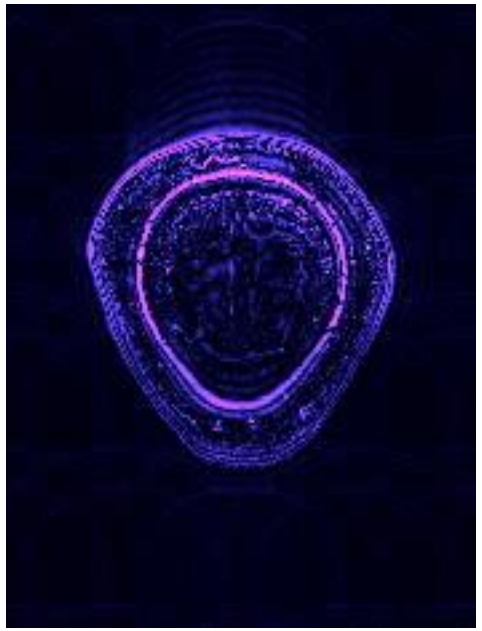
Ground Truth

Zero-Filled

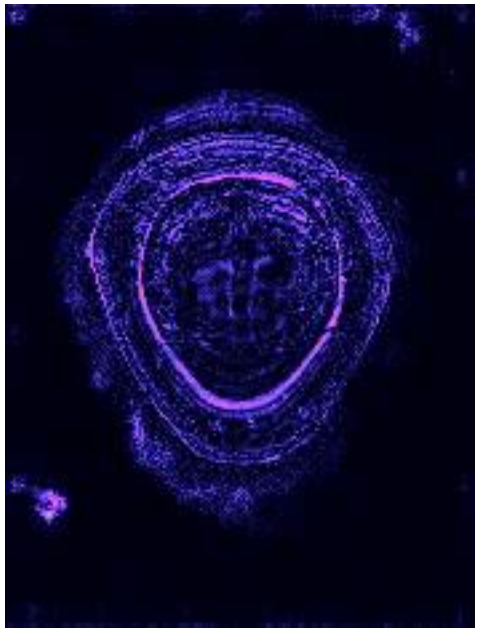


Output

Input Error Map



Output Error Map

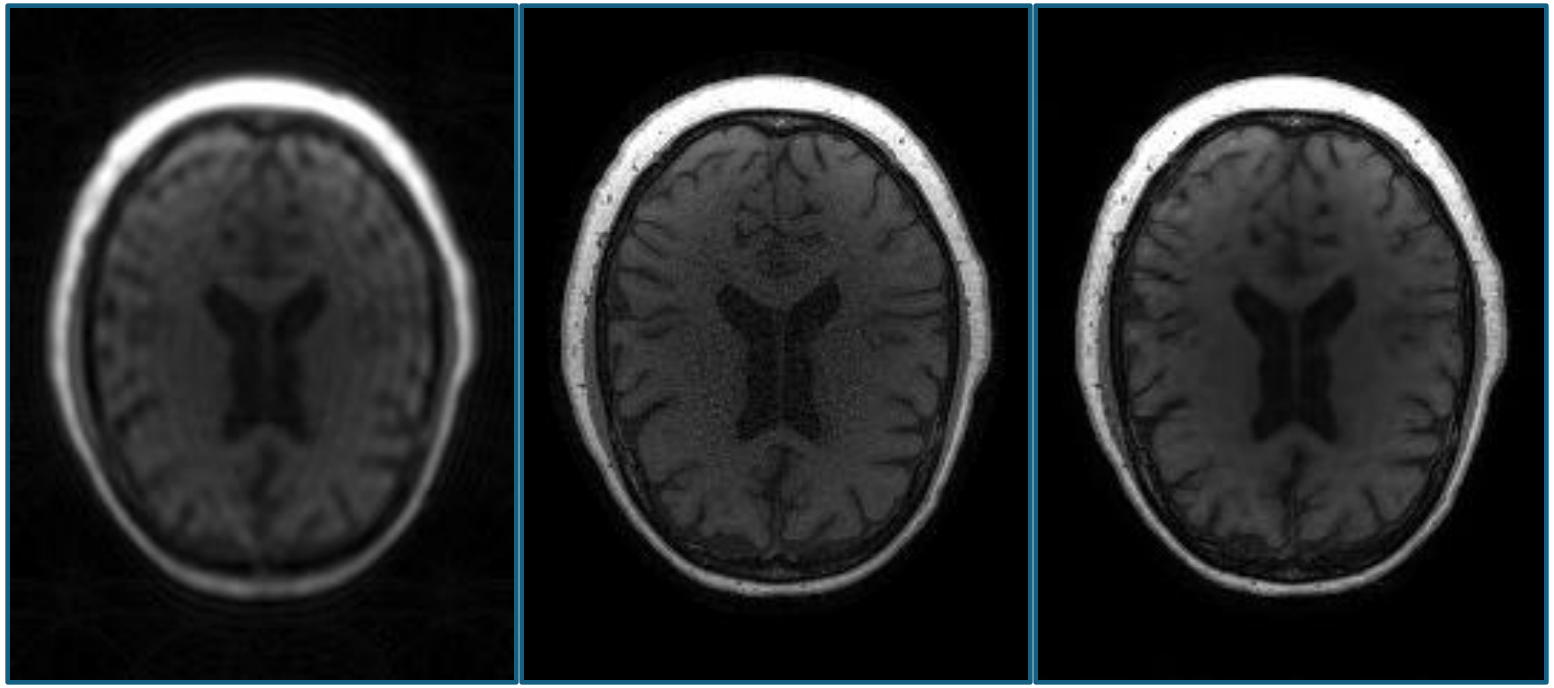


Comparisons

Ground Truth

Real 2x

Zero-Fill **SENSE** **Ours**



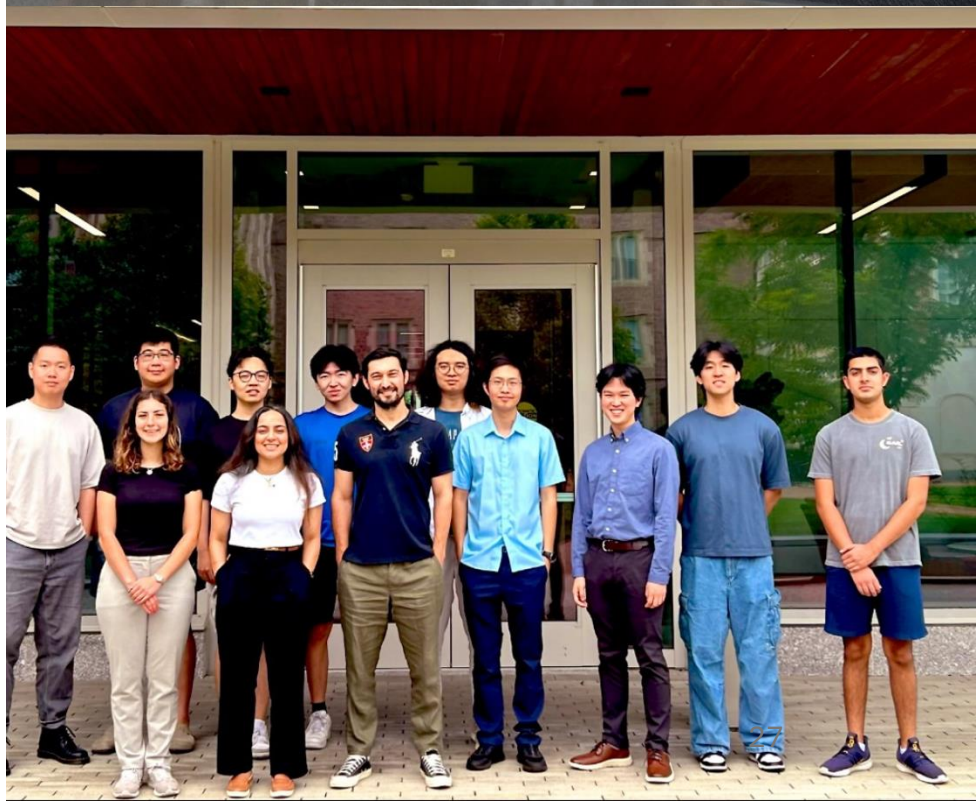
Real 4x

Future Work

1. Bigger Model => better performance?
2. Replacement for U-Net
3. Register 2x and 4x data (physical shifting) for one-to-one mapping

Acknowledgements

1. WUSEF/Thermal Management REU/CURB
2. Janie Brennan PhD, Sharniece Holland-Trogler PhD, Emily Boyd PhD
3. Computational Imaging Group
4. Ulugbek S. Kamilov PhD, Shirin Shoushtari
5. Computer Science REU, Dr. Shook
6. Last Mile Education



Extra Slides

2D Fourier transform

Definition

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-j2\pi(ux+vy)}\,dx\,dy,$$
$$f(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v)e^{j2\pi(ux+vy)}\,du\,dv$$

where u and v are spatial frequencies.

Also will write FT pairs as $f(x,y) \Leftrightarrow F(u,v)$.

- $F(u,v)$ is complex in general,

$$F(u,v) = F_{\mathrm{R}}(u,v) + jF_{\mathrm{I}}(u,v)$$

- $|F(u,v)|$ is the magnitude spectrum
- $\arctan(F_{\mathrm{I}}(u,v)/F_{\mathrm{R}}(u,v))$ is the phase angle spectrum.
- Conjugacy: $f^*(x,y) \Leftrightarrow F(-u,-v)$
- Symmetry: $f(x,y)$ is even if $f(x,y) = f(-x,-y)$

Mean Squared Error (MSE)

The MSE loss measures the average of the squares of the errors—that is, the average squared difference between the estimated values and the actual value.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Case 2: `penalty = 'l2'`

This calculates the **squared L2 norm** of the image gradient. It penalizes the sum of *squared* differences, which punishes large, sharp changes much more heavily than small ones.

$$L_{TV_{L2}}(I) = \frac{1}{2}\left(\mathbb{E}[(\nabla_x I)^2] + \mathbb{E}[(\nabla_y I)^2]\right)$$

Breakdown of the Equation:

The components are the same as above, with one key difference:

- $(\cdot)^2$: The square of the gradient values. This corresponds to the lines `dy = dy * dy` and `dx = dx * dx` inside your `if` block.

This equation represents the calculation when the `penalty` is set to `'l2'`.

Extra Slides

Definition [edit]

PSNR is most easily defined via the mean squared error (*MSE*). Given a noise-free *m*×*n* monochrome image *I* and its noisy approximation *K*, *MSE* is defined as

$$MSE = \frac{1}{m\,n}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[I(i,j) - K(i,j)]^2.$$

The PSNR (in dB) is defined as

$$\begin{aligned} PSNR &= 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right) \\ &= 20 \cdot \log_{10}\left(\frac{MAX_I}{\sqrt{MSE}}\right) \\ &= 20 \cdot \log_{10}(MAX_I) - 10 \cdot \log_{10}(MSE). \end{aligned}$$

Here, MAX_I is the maximum possible pixel value of the image. When the pixels are represented using 8 bits per sample, this is 255. More generally, when samples are represented using linear PCM with *B* bits per sample, MAX_I is $2^B - 1$.

General formula and components [edit]

The SSIM formula is based on three comparison measurements between the samples of x and y: luminance (l), contrast (c), and structure (s). The individual comparison functions are:[4]

$$l(x,y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}$$

$$c(x,y) = \frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}$$

$$s(x,y) = \frac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3}$$

The SSIM for each block is then a weighted combination of those comparative measures:

$$\mathrm{SSIM}(x,y) = l(x,y)^\alpha \cdot c(x,y)^\beta \cdot s(x,y)^\gamma$$

Choosing the third denominator stabilizing constant as:

- $c_3 = c_2/2$

leads to a simplification when combining the *c* and *s* components with equal exponents ($\beta = \gamma$), as the numerator of *c* is then twice the denominator of *s*, leading to a cancellation leaving just a 2. Setting the weights (exponents) α, β, γ to 1, the formula can then be reduced to the special case shown above.

https://en.wikipedia.org/wiki/Peak_signal-to-noise_ratio
https://en.wikipedia.org/wiki/Structural_similarity_index_measure